WORKSHEET

Chapter 9: Sample proportions

1 A sample of 150 items on a production line in a factory is inspected for faults. Twenty-five were found to have some kind of fault. What is the sample proportion of faulty items on this production line?

2 A pair of dice is rolled and the total of their faces noted. If the dice are rolled 80 times and a total of 6 occurs 14 times, calculate the sample proportion of a total of 6 when a pair of dice is rolled.

3 When 30 students at a large school were surveyed, it was found that 8 students had 4 people in their household. What is the sample proportion of students who come from a four-person household?

4 From a sample of 30 students at a Melbourne school, it was found that 9 came from situations where total household income exceeded $100 000 p.a.

a Estimate the probability that an Australian household has an annual income in excess of $100 000.

b State any problems with treating this as a reliable estimate.

5 A card is selected from a standard deck of playing cards and then replaced. This is repeated 120 times. It was found that on 8 occasions, the card was a heart face card. A face card is a J, Q or K.

a What is the estimated probability of drawing a heart face card from a standard deck of playing cards?

b What is the actual probability of drawing a heart face card from a standard deck of playing cards?

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6 A pair of dice is rolled and the numbers on the upper faces are noted. The dice are rolled 110 times and a 5 appears on at least one die 35 times.

a Estimate the probability a 5 appears on at least one die when a pair of dice is rolled.

b What is the actual probability of a 5 appearing on at least one die when a pair of dice is rolled?

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Answers

1 $\frac{1}{6}$

2 $\frac{7}{40}$

3 $\frac{2}{15}$

4 **a** $\frac{3}{10}$

b Many problems including: the school may have drawn students from high socioeconomic areas, Melbourne incomes may be different from the remainder of Australia, etc.

5 **a** $\frac{1}{15} \approx 0.06667$

b $\frac{3}{52} \approx 0.05769$

6 **a** $\frac{7}{22} \approx 0.3182$

b $\frac{11}{36} \approx 0.3056$

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